PROJECT 12 · PHASE 2

Smart Waste Segregation

Transfer learning, a hybrid model and one frozen test

SIX CATEGORIES

- Cardboard
- Glass
- Metal
- Paper
- Plastic
- Trash

TEAM

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WHERE WE STARTED

Phase 1 left a 0.8999 baseline to beat

0.8999

Phase 1 test macro F1

Handcrafted colour and texture features with a calibrated RBF SVM.

6

Metal photos called glass

The largest single confusion on the Phase 1 test set.

Same TrashNet data, same frozen train / validation / test splits and folds as Phase 1, so every Phase 2 number is directly comparable.

Phase 2 asks

- Does an ImageNet-pretrained CNN beat handcrafted features?
- Does combining the two do better than either alone?
- Does the answer hold on 377 unseen test images?

Rule we kept

Decisions on training folds; calibration and thresholds on validation; the test set used once.

THE PIPELINE

Seven steps, one frozen protocol

01
Data
Frozen splits, train-only augmentation
Om

02
Backbones
MobileNetV2 and EfficientNet-B0
Aditya

03
Train
Head first, then last two blocks
Shourya

04
Ablate
13 pre-registered settings, 65 runs
Aditya

05
Combine
Three hybrid designs
Daksh

06
Freeze
Calibrate, fix thresholds, test once
Prajjwal

07
Demo
Model selector with abstention
Saksham

Which data decides what

1,764 training images in 5 frozen folds: every architecture, ablation and hybrid choice.

380 validation images: calibration checks and abstention thresholds only.

377 test images: one frozen run at the very end.

DATA AND REGULARISATION

Augmentation only where the model learns

Random crop
80–100% of the image

Horizontal flip
p = 0.5

Rotation
± 10°

Colour jitter
brightness and contrast 0.15, saturation 0.1

Same preprocessing as Phase 1
EXIF rotation, white letterbox to 224 × 224, ImageNet normalisation. Validation and test images are never augmented, and near-duplicates stay in one split.

4.33

times as many paper photos (416) as trash photos (96) in training

Two fixes, both tested

- Class-weighted loss
- Balanced sampling

Both run as ablations rather than assumed to help.

ARCHITECTURES

Two light ImageNet backbones

Selected

MobileNetV2

2,223,872
backbone parameters

- Depthwise-separable convolutions
- Inverted residuals with linear bottlenecks
- 1,280-number image summary

Compared

EfficientNet-B0

4,007,548
backbone parameters

- MBConv blocks with squeeze-and-excitation
- Compound scaling of depth, width, resolution
- 1,280-number image summary

Why pretrained

ImageNet features (edges, textures, shapes) transfer well, so 1,764 training photos are enough to adapt them.

No centre crop

A letterbox keeps the whole object in view instead of cutting its edges.

TRAINING

Train the head, then the last two blocks

Head
Dropout 0.2 → Linear 1,280 → 6

Optimiser
AdamW, weight decay 0.01

Learning rates
Head 1e-3 · unfrozen blocks 1e-4

Schedule
Cosine to 1e-5 over 40 epochs

Early stopping
Patience 8 on held-out fold loss

Final model
Retrained on all 1,764 for 13 epochs

Frozen means frozen

Even with weights frozen, BatchNorm statistics and EfficientNet's stochastic depth still change in training mode. Frozen blocks are kept in evaluation mode, and a test checks nothing moves.

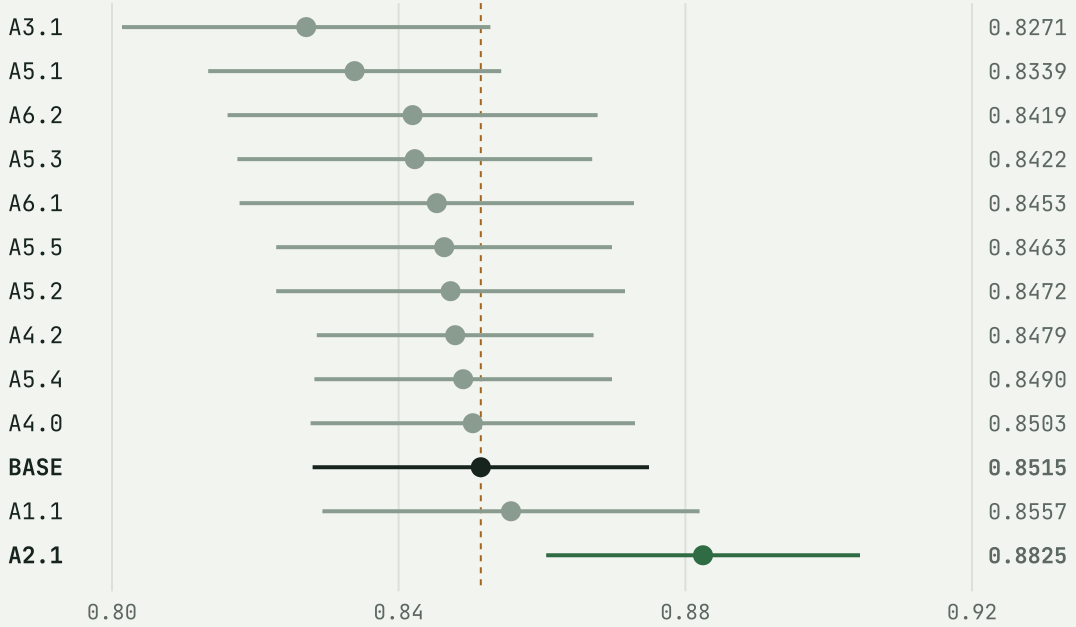
Same schedule at the end

The final 13-epoch run follows the 40-epoch cosine curve exactly as the cross-validation runs did.

ABLATION STUDY

Only fine-tuning clearly beat the baseline

MEAN MACRO F1 (DOT) ± STD (LINE), 5 TRAINING FOLDS · DASHED: BASE



SELECTED · A2.1

0.8825

± 0.0219, last two MobileNetV2 blocks fine-tuned

Gain over BASE **+0.0310**, larger than the fold spread **0.0235**: the only clear winner.

893,766 trainable parameters · median best epoch **13**

WHAT THE ABLATIONS TAUGHT US

Small changes mostly sat inside the noise

No augmentation **-0.0176**

Within the fold spread: no clear difference.

160 × 160 input **-0.0244**

Just beyond the spread: smaller images hurt.

EfficientNet-B0 **+0.0042**

No clear gain for 1.8× the parameters.

Class-weighted loss **-0.0061**

No clear F1 change; trash recall 65.6% → 72.9% (mean per fold).

Caveat we report. Frozen models reached their best epoch at 38–40 of the 40-epoch cap (BASE median 38), while A2.1 stopped by itself near 20. Part of A2.1's lead may come from that budget; the cap was pre-registered, so the selection stands.

HYBRID MODELS

Three hybrid designs; late fusion leads

H1

ML on CNN features

An SVM on the 1,280 EfficientNet-B0 features.

0.8788

± 0.0185 CV macro F1

H2

Early fusion

Handcrafted and CNN features joined, each block scaled, one SVM.

0.9175

± 0.0262 CV macro F1

H3

Late fusion

Average the probabilities of the Phase 1 SVM and the A2.1 CNN.

0.9364

± 0.0171 CV macro F1

Single models on the same five folds: Phase 1 SVM **0.8844** · A2.1 CNN **0.8825**. Rationale: handcrafted colour and texture and CNN shape features fail on different photos.

WHY LATE FUSION WORKS

The two models make different mistakes

295

training photos where the SVM and CNN disagree

16.7% of 1,764

46

photos where both are wrong

2.6% of 1,764: different mistakes, so averaging fixes most

The weight is chosen on the other four folds every time, so no fold helps pick its own weight. All five folds chose `w = 0.5`.

Pre-registered rule

A hybrid wins only if it beats the best single model by more than the fold spread.

+0.0539

H3 over A2.1, against σ = 0.0219: a clear win in cross-validation.

H3 vs H2 (+0.0189) is inside the noise; H3 goes forward as the highest mean.

CALIBRATION AND FREEZE

Everything fixed before the test set opened

Fixed on training and validation data

Model	Temperature	Headline	Demo
Phase 1 SVM	0.834	0.45	0.76
A2.1 CNN	1.103	0.50	0.87
H3 hybrid	0.536	0.00	0.53

Temperature fitted on out-of-fold training predictions. Headline threshold: highest coverage with ≥ 90% accuracy on validation. Demo threshold: ≥ 95% accuracy at ≥ 80% coverage.

Frozen before the test

Model file hashes, temperatures and both thresholds were committed in `protocol.json` (commit `16e4b9f`) before a single test image was scored.

The code refuses to touch the test set without that file and refuses to run the test twice.

Test images never tuned anything.

FROZEN TEST RESULT

One measurement on 377 held-out images

0.9233

H3 hybrid · test macro F1

95% bootstrap interval 0.889–0.952, on 377 images it never saw

93.63%

Accuracy, all 377

353 correct

96.55%

Coverage at 0.53

364 answered, 13 “not sure”

95.33%

Accepted accuracy

347 of 364 answered photos right

Best calibrated of the three models: expected calibration error 0.013.

TEST COMPARISON

Best F1, accuracy and calibration: the hybrid

Model	Macro F1	95% CI	Accuracy	ECE	Coverage	Accepted acc.
Phase 1 SVM	0.8963	0.860–0.926	90.19%	0.036	84.4%	97.2%
A2.1 CNN	0.8577	0.813–0.897	88.06%	0.037	76.4%	95.5%
H3 hybrid	**0.9233**	**0.889–0.952**	**93.63%**	**0.013**	**96.6%**	**95.3%**

Coverage and accepted accuracy at each model's frozen demo threshold · 377 test images

Most useful answers

At their demo thresholds H3 answers 96.6% of photos, the SVM 84.4% (more accurate when it answers), the CNN 76.4%.

CNN alone is weakest

A2.1 scores 0.8577, below the SVM's 0.8963, but the gap is not significant.

Phase 1 reproduced

The refit SVM matches Phase 1's 90.19% accuracy; F1 differs (0.8963 vs 0.8999) only through calibration.

SIGNIFICANCE

A sure win over the CNN, likely over the SVM

H3 vs A2.1 only one right: 23 vs 2		p = 0.00002	significant
H3 vs SVM only one right: 23 vs 10		p = 0.035	suggestive only
A2.1 vs SVM only one right: 25 vs 33		p = 0.358	no difference

Exact McNemar tests on the photos where only one model is right. With three comparisons the bar is **p < 0.0167** (Bonferroni). H3 clearly beats the CNN; its lead over the SVM (p = 0.035) is suggestive, not proven.

ERROR ANALYSIS

Fewer look-alike mistakes than Phase 1

True \ Pred.	Cardbd.	Glass	Metal	Paper	Plastic	Trash
Cardboard	60	0	0	0	0	0
Glass	0	70	1	0	3	1
Metal	1	2	58	0	0	0
Paper	1	0	1	84	0	3
Plastic	0	4	1	2	63	2
Trash	0	0	1	1	0	18

H3 hybrid on 377 test images · rows: true class · green diagonal = correct (353 of 377)

4

plastic photos called glass, now the largest confusion

2

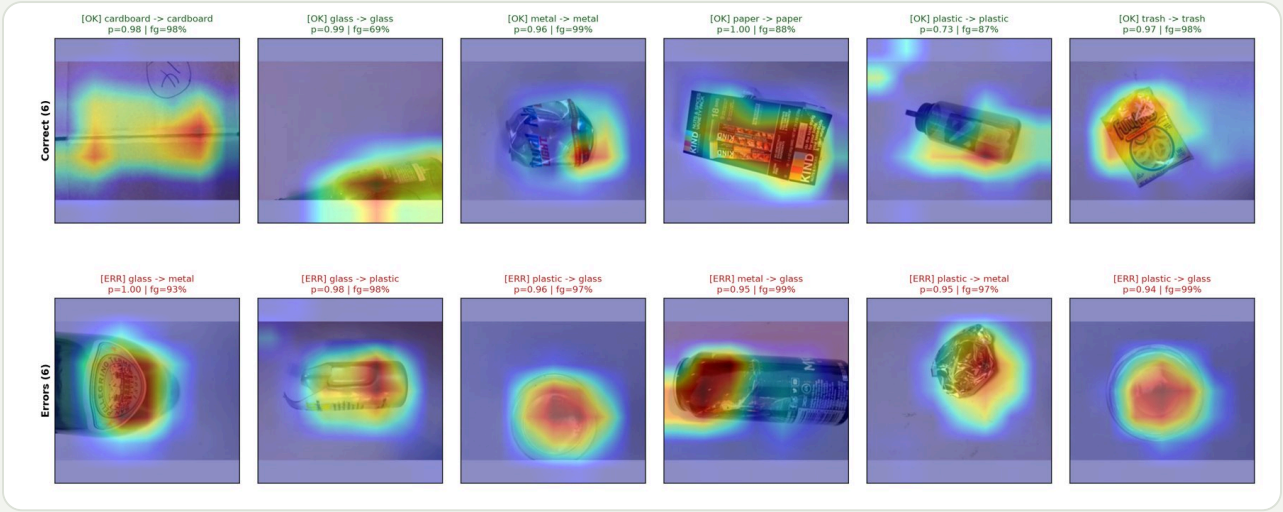
metal photos called glass; the Phase 1 model made 6

87.50%

plastic recall (63 of 72), the lowest of any class

GRAD-CAM · A2.1 CNN

Where the CNN looks: 6 correct, 6 wrong



Looks at the object

In 11 of 12 maps the object gets more attention than its share of the image.

Not an explanation

Errors focus on the object too; 12 maps cannot explain score gaps.

DEMO

Pick a model, upload a photo



Metal · 99.9%
`threshold 0.53`

`answered`



Cardboard 52.5% · paper 47.5%
`below 0.53`

`not sure`

Real H3 outputs from the merged demo code on two validation photos (not test photos).

1. **Pick a model**
   H3 hybrid (recommended), A2.1 CNN or Phase 1 SVM
2. **Upload a photo**
   Local Gradio app; public sharing is off
3. **Check the files**
   Every model file's SHA-256 must match protocol.json
4. **Read six probabilities**
   Temperature-scaled with the frozen value
5. **Flag uncertainty**
   Below the frozen demo threshold, say “not sure”

LIMITATIONS

What these results do not show

Cross-validation is optimistic

Settings were chosen on the same five folds; the test run is the fair number.

Small test set

377 images, only 20 of them trash: wide confidence intervals.

One photo setting

TrashNet photos sit on a white board; our everyday-photo set was not collected.

Thresholds drift

A2.1's demo threshold covered 76.4% of test photos vs 80% on validation.

GPU runs vary slightly

Retraining gives a new model and new hashes; results are frozen by hash.

Not disposal advice

Labels are materials; local recycling rules decide what goes where.

THE TEAM

Six contributions, one frozen evaluation

Om Yadav
Data pipeline
Splits, augmentation, sampling

Aditya Bhardwaj
Backbones and ablations
Embeddings, 13-setting study

Shourya Bafna
Training loop
Fine-tuning and final CNN

Daksh Batra
Hybrid models
Early and late fusion

Prajjwal Tripathi
Calibration and test
Frozen protocol, statistics

Saksham Miglani
Demo and report
Model selector, IEEE paper

THANK YOU

Questions

A hybrid that beats Phase 1, measured once and honestly

0.9233
H3 test macro F1

13
pre-registered ablations

p < 0.0001
H3 vs CNN, McNemar

96.6%
answered, 95.3% of them right

Code and records: github.com/shourya2006/smart-waste-segregation

BACKUP

References

Sandler et al. (2018)
MobileNetV2

Yosinski et al. (2014)
Transferability of features

Loshchilov and Hutter (2017)
SGDR, cosine schedule

Huang et al. (2016)
Stochastic depth

Guo et al. (2017)
Calibration of neural networks

Tan and Le (2019)
EfficientNet

Loshchilov and Hutter (2019)
AdamW

Ioffe and Szegedy (2015)
Batch normalisation

Wolpert (1992)
Stacked generalisation

Selvaraju et al. (2017)
Grad-CAM

Project metrics come from the versioned CSV and JSON records in the repository, not from these papers.

BACKUP

How to reproduce

Check the frozen results

Full steps: `docs/phase-2/reproduction.md`

- Install from the pinned lock files
- `waste-data download` (1,764 / 380 / 377)
- Unzip the two frozen models; hashes are checked on load
- `notebooks/phase2-summary.ipynb` rebuilds the tables
- `python app.py` runs the demo

Frozen file hashes

A2.1 `model.pt`

```
26592f8b5aee95eda28ef6088af8c7d737e7e6b9d38a6
733c9faa5c1b4c5d28e
```

Hybrid `svm_pipeline.joblib`

```
ca5a5411db8fbbe79296e552e50f7340fe3c376314869
7151a83cc858f5d080a
```